18 A scale model of a water tank consists of a cone mounted on a cylinder. The cylinder and the cone have the same radius of 10.5 cm. The volume of the model is 4620 cm^3.

$\left(\text{Take } \pi = \frac{22}{7}\right)$

(a) Given that the height of the cylindrical part is 12 cm, determine the height of the conical part. (3 marks)

(b) Calculate the surface area of the model. (5 marks)

(c) If the volume of the actual tank is 231 m^3, calculate the radius of the cylindrical part of the actual tank. (2 marks)

(b) Estimate the median mark, correct to 1 decimal place. (2 mark

6 A student stands 50 m from the foot of a building on a level ground. The eye level of the student is at a height of 1.66 m. She observes the top of the building at an angle of elevation 30°. Calculate the height of the building. (3 mar

2

7 Quadrilateral ABCD is such that $\angle BAD = 100°$ and $\angle ABC = \angle ADC = 90°$. Diagonal $AC = 10$ cm is a line of symmetry to the quadrilateral. Calculate the area of quadrilateral ABCD. (3 ma

3

8 A Canadian tourist visited Kenya. She had 4 800 Canadian dollars to spend during her visit. She converted all the Canadian dollars to Kenyan shillings at the exchange rate shown.

	Buying (Ksh)	Selling (Ksh)
1 Canadian Dollar	118.26	122.31

The tourist spent Ksh 346 000 during the visit. She converted the balance to Canadian dollars on her departure from Kenya. Calculate the amount she got in Canadian dollars.

(3 marks)

9 The position vectors of points X, Y and Z are $\mathbf{x} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$, $\mathbf{y} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\mathbf{z} = \begin{pmatrix} 12 \\ 10 \end{pmatrix}$ respectively.

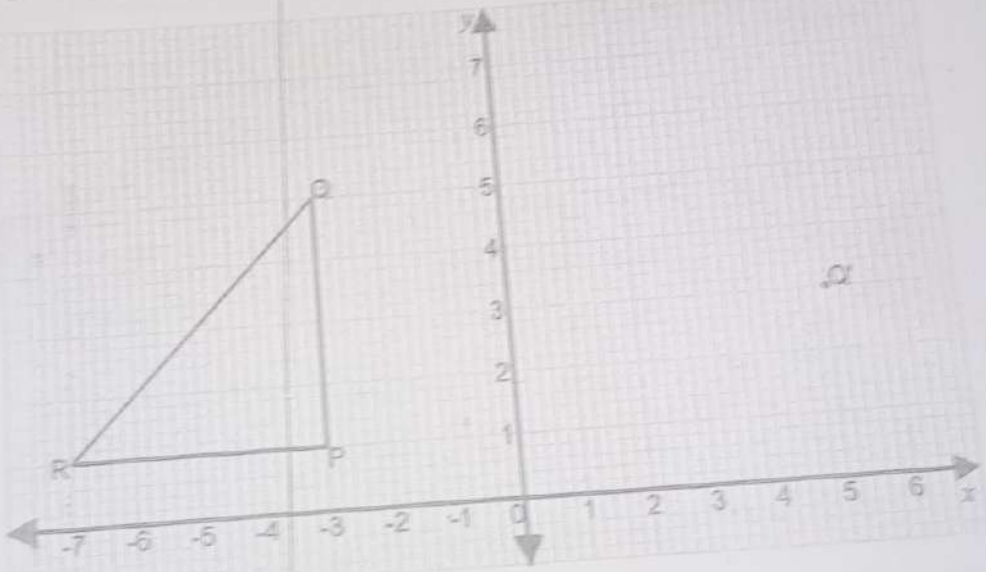
Find the value of scalars a and b such that $a\mathbf{x} + b\mathbf{y} = \mathbf{z}$.

(4 marks)

10 Given that $\cos(\theta + 20) = \sin 37.5°$, find the value of θ in degrees.

(2 marks)

16 The following figure shows triangle with vertices P(–3, 1), Q(–3, 5) and R(–7, 1). Point Q'(5, 3) is the image of Q(–3, 5) under a rotation about the origin O.



1 (a) Determine the angle of rotation. (1 m

13 (b) Complete $\Delta PQR'$ the image of ΔPQR under the rotation. (2 m

11 Factorise $(4a^2 - b^2)$. Hence use the factorisation to evaluate $(4 \times 10^2 - 9^2)$. (3 marks)

12 The base of a right pyramid of height 12 cm is a square of length 10 cm. Calculate the surface area of the pyramid. (4 marks)

13 A school intended to buy 5 more chairs than tables. Each chair costs Ksh 600 while each table costs Ksh 5 000. The school is allowed to use at most Ksh 50 000 for this purpose. Find the maximum number of chairs that the school can buy. (3 marks)

14 The curve of the function $y = x^2 - 4x + 4$ passes through the point P (3, 1). Find the equation of a tangent to the curve at P. (4 marks)

15 The matrix $\begin{pmatrix} x & 6 \\ 2 & 3x \end{pmatrix}$ is a singular matrix. Find the possible values of x. (2 marks)

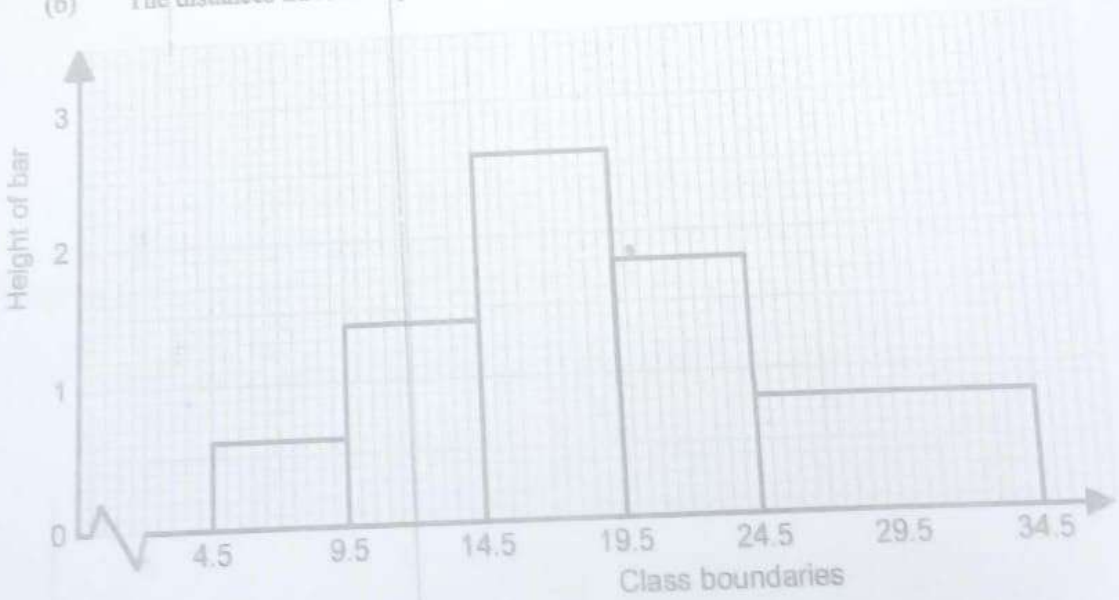
19 The following table shows the distances in kilometres travelled by 40 trainees attending a workshop.

Distance (km)	5 - 9	10 - 14	15 - 19	20 - 24	25 - 34
No. of trainees	3	7	13	9	8

(a) (i) State the modal frequency. (1 mark)

(ii) Calculate the mean distance. (4 marks)

(b) The distances travelled by trainees are represented on a histogram as shown.



8 Use the histogram to:

(i) estimate the median distance; (3 marks)

(ii) determine the number of trainees who travelled 27 km or more. (2 marks)

SECTION II (50 marks)

Answer only five questions in this section in the spaces provided.

17 Members of a Women's group harvested and packed honey into 2 types of jars; type P and type Q. There were 90 more of type P jars than type Q.

(a) (i) Given that the number of type P jars and that of type Q jars were in the ratio 5 : 2, determine the number of type P jars. (2 marks)

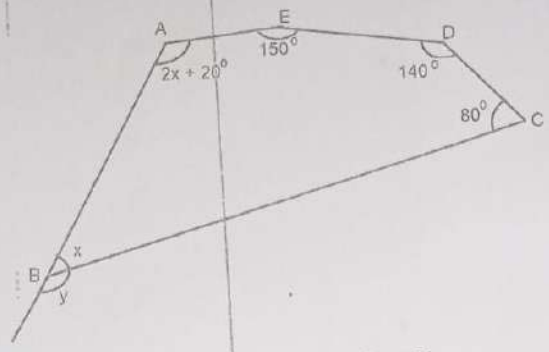
(ii) The members put 500 g of honey into each of type P jar and 750 g of honey into each of type Q jar.
Determine the total mass of honey, in kg that was packed. (3 marks)

(b) Type P jars were packed into cartons, each containing 15 jars. Each carton of the type P jars was sold for Ksh 7 125. Type Q jars were packed, into cartons, each containing 12 jars. Each carton of type Q jars was sold for Ksh 8 550.

(i) Calculate the amount of money raised by the group from the sale of honey. (2 marks)

(ii) Given that the Women's group used a total of Ksh 75 000 in production of the honey, calculate the % profit made by the group. (3 marks)

4 The following figure shows an irregular pentagon ABCDE in which $\angle A = 2x + 20°$, $\angle B = x$, $\angle C = 80°$, $\angle D = 140°$ and $\angle E = 150°$.



Determine the value of y, the exterior angle at B. (3 marks)

5 The scores obtained in a Mathematics test by a group of students were recorded as shown in the following frequency distribution table.

Marks	Number of Students	Cumulative Frequency
11-20	1	1
21-30	3	4
31-40	4	8
41-50	8	16
51-60	12	28
61-70	x	34
71-80	3	37
81-90	2	39
91-100	1	40

(a) Determine the number of students who scored 61-70 marks. (1 mark)

SECTION I (50 marks)

Answer all the questions in this section in the spaces provided.

1 Use logarithm tables to evaluate. (4 marks)

$$\frac{10.32^2 \times \sqrt{0.6318}}{427}$$

2 Given $\frac{3x-2y}{x} = \frac{4}{5}$, find the ratio $x : y$. (3 marks)

3 Solve for x in the equation. (3 marks)

$$25^{x^2} = 125^{12-2x}$$

(a) Determine the number of students who scored 61-70 marks. (1 mark)